# Youngseok Lee

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## Education

**Sungkyunkwan University**, Nano Engineering

## Experience

**NORMA Inc.**, Quantum AI Developer, Quantum AI Team
Research and development of quantum machine learning models and the classical simulation tools used to train them.

- Co-developed and maintain PADO-Pauli, a GPU-accelerated differentiable Pauli-propagation surrogate engine (Python package padopauli).
- Proposed and analyzed mixed IQP-QCBMs, ancilla-extended IQP generative models with provable barren-plateau avoidance and a cluster initialization scheme.
- Developed SAFE ma-QAOA, a surrogate-assisted, parameter-distilled multi-angle QAOA pipeline with exact fine-tuning.
- Applied QGAN and IQP-QCBM models to molecular generation for KRAS G12D inhibitor candidates, including experiments on Quantinuum H2 and IonQ Forte.
- Contributed to an in-house quantum SDK (circuit representation, user API, circuit output and visualization).

## Publications

**Trainability and Mode Separation of Mixed IQP-QCBMs**
Youngseok Lee, Hyunwoo Kim
arxiv.org/abs/2607.27883

**SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation**
Youngseok Lee, Hyunwoo Kim

**PADO-Pauli: A GPU-Accelerated Implementation of Pauli Propagation**
Hyunwoo Kim, Youngseok Lee
github.com/Norma-Q/PADO-Pauli

## Talks

- **SAFE ma-QAOA**, contributed talk (accepted), IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026.
- **PADO-Pauli**, poster (accepted), IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026.
- **Trainability and Mode Separation of Mixed IQP Circuits**, contributed talk, ML4QT Symposium, Institute for Quantum Computing, University of Waterloo, September 23–25, 2026.

## Patents

- **Quantum Computing Method Using a Pauli Graph Generated by Zero Filtering to Perform Surrogate Operations and Quantum Computing System Performing the Same.** Korean patent, application no. 10-2026-0070495 (filed 2026-04-20), registered. Applicant: NORMA Inc.
- **Quantum Artificial Intelligence Training Method Performing Surrogate Training Using Pauli Graphs and Quantum Artificial Intelligence Training System Performing the Same.** Korean patent, application no. 10-2026-0070494 (filed 2026-04-20), registered. Applicant: NORMA Inc.

## Research Mentoring

- Supervised research intern Dohyun Kim (January 19 – April 19, 2026) on *Reinforcement Learning for Adaptive Pauli Propagation*, and wrote a letter of recommendation.

## Skills

**Quantum computing**

**Software**

## Interests

**Research interests**